

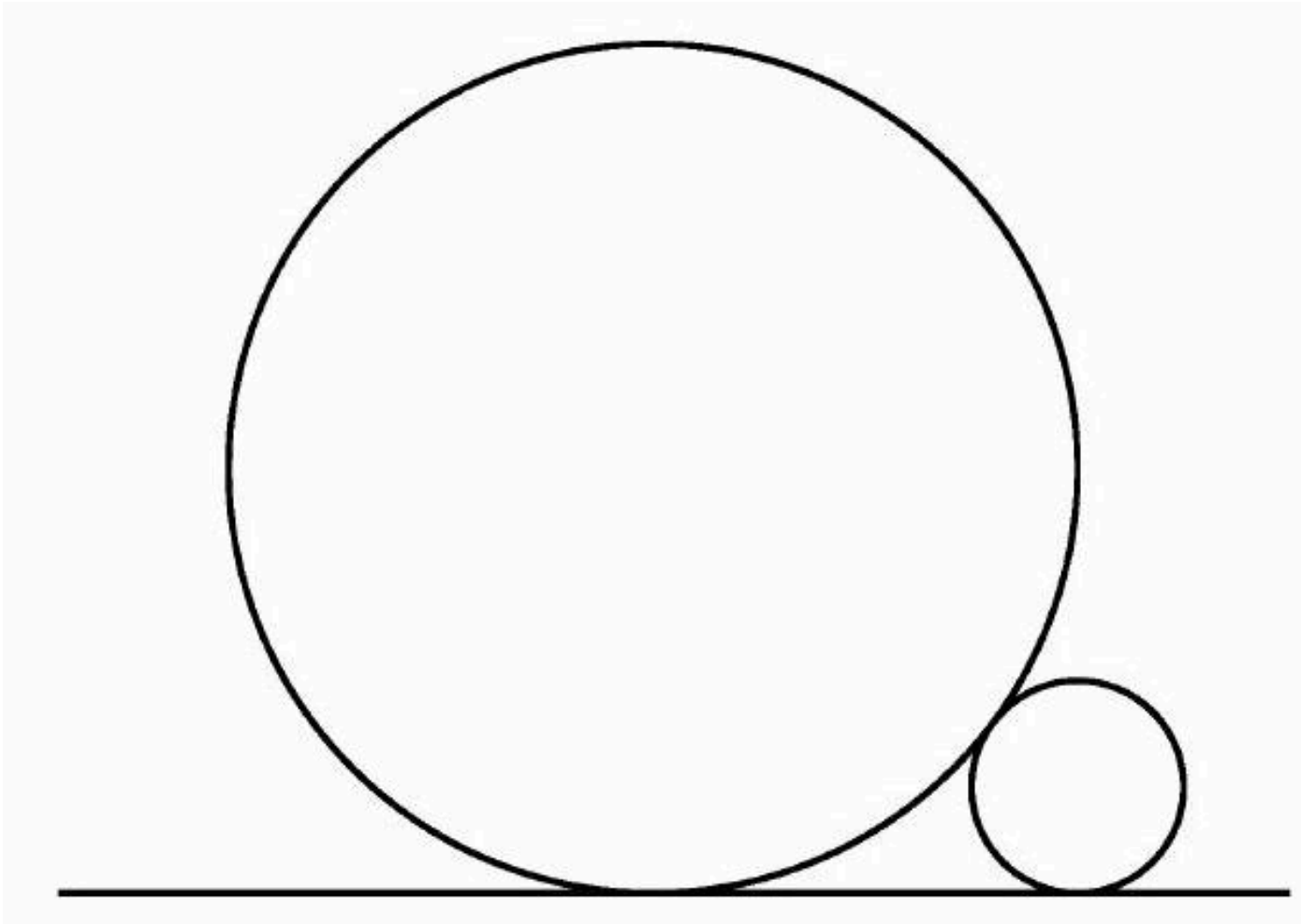
2024 AMC 10B Problem 21 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 21. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 21

Problem:

Two straight pipes (circular cylinders), with radii 1 and $\frac{1}{4}$, lie parallel and in contact on a flat floor. The figure below shows a head-on view. What is the sum of the possible radii of a third parallel pipe lying on the same floor and in contact with both?



Answer Choices:

A. $\frac{1}{9}$

B. 1

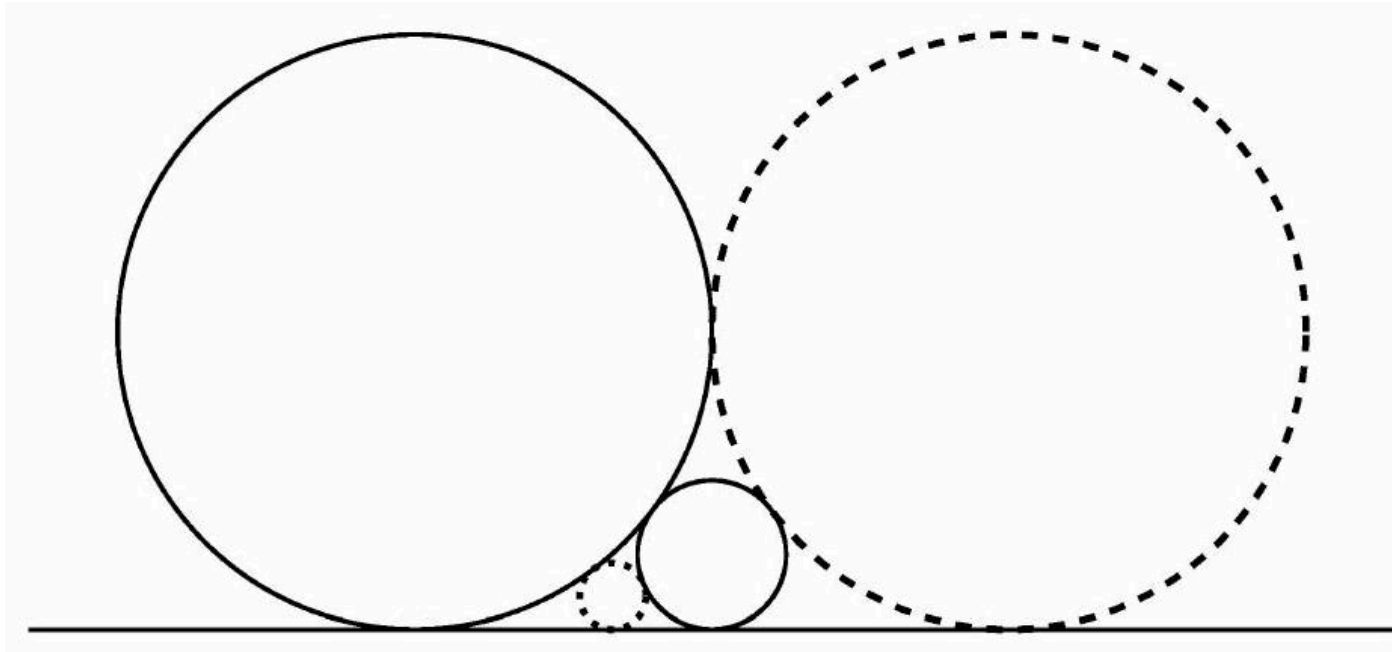
C. $\frac{10}{9}$

D. $\frac{11}{9}$

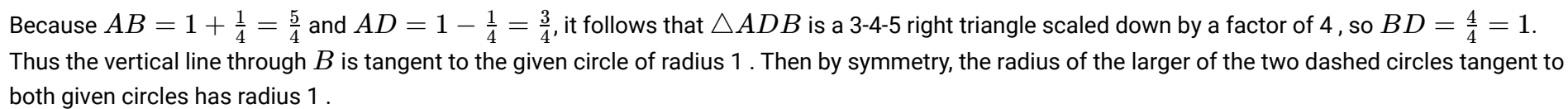
E. $\frac{19}{9}$

Solution:

There are two possible positions for the third pipe-either nestled in the gap between the pipes or outside. See the figure below.



Consider the blown-up figure below. In this diagram, A is the center of the circle of radius 1, B is the center of the circle of radius $\frac{1}{4}$, and C is the center of the third circle nestled in the gap. The horizontal lines through B and C intersect the vertical line through A at D and E , respectively, and F is the foot of the perpendicular from C to $\overline{BD}$.


$$(1-x)^2 + \left(1 - \frac{3}{4} - r\right)^2 = \left(\frac{1}{4} + r\right)^2$$

which simplifies to $(1 - x)^2 = r$. Combining these equations gives $x^2 = 4(1 - x)^2$, which is equivalent to $3x^2 - 8x + 4 = 0$ and $(3x - 2)(x - 2) = 0$. Because $x < 1$, the relevant solution is $x = \frac{2}{3}$, and the radius of the smaller circle is $\frac{1}{4} \cdot \left(\frac{2}{3}\right)^2 = \frac{1}{9}$.

The requested sum of possible radii is $1 + \frac{1}{9} = (C) \boxed{\frac{10}{9}}$.

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) 

Powered by [Wiki.js](#)